

AD-GS: Alternating Densityfication for Sparse Input 3D Gaussian Splatting



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gurutvapatile.github.io/publications/2025/ADGS.html

① 3DGS with Sparse-Input

- 3DGS performs well with dense views.
- Real-world capture often yields few input views (3–24).

Train Renders (3 views)



Novel Views



- Issues: floaters, inaccurate geometry.
- Due to overfitting.

② Prior Work on Sparse-Input 3DGS

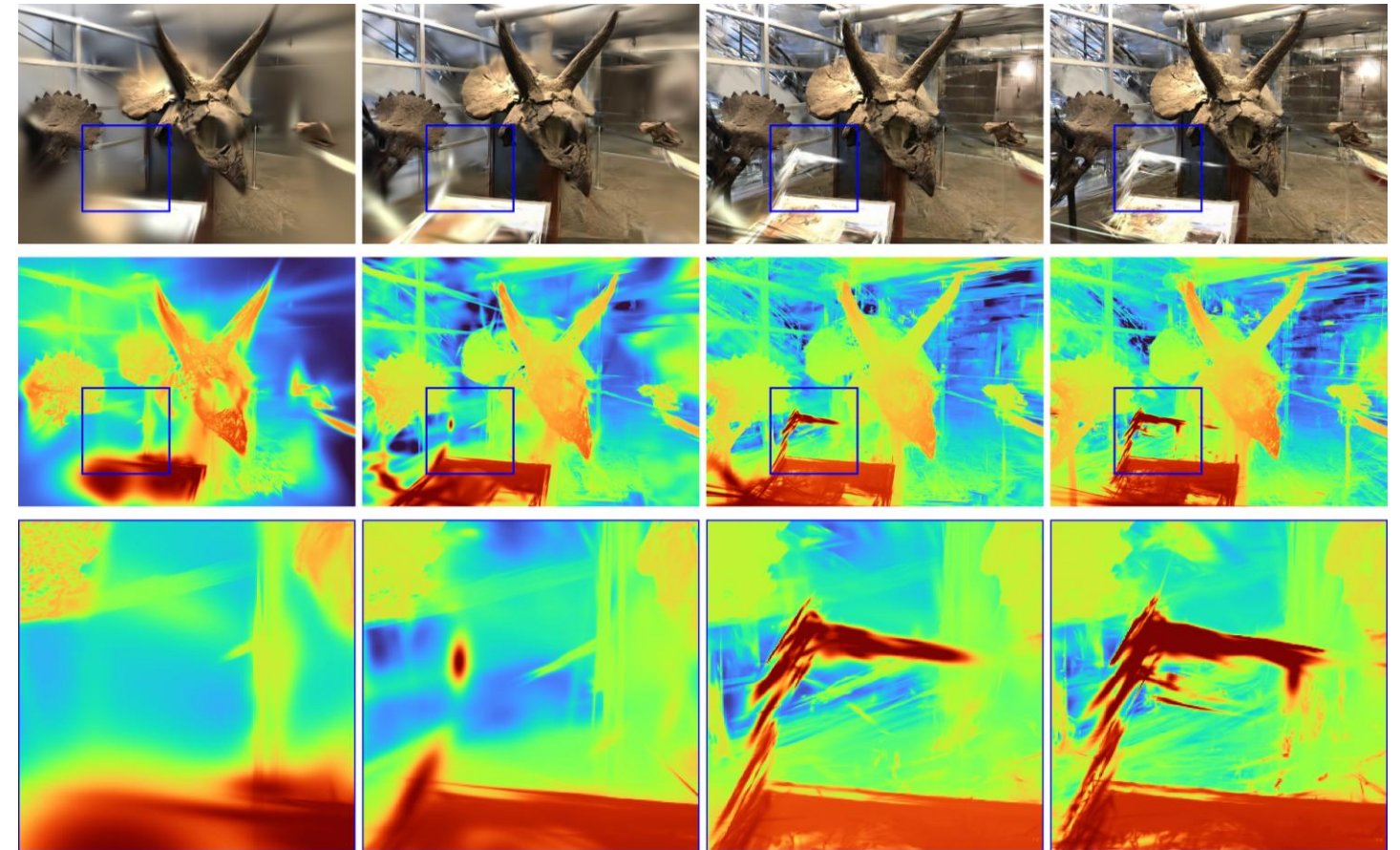
- FSGS – uses priors from deep depth prediction networks, but such priors may not generalize.
- CoR-GS – enforces consistency between paired 3DGS models, but this smooths reconstructions.
- DropGaussian – randomly drops Gaussians during training, but this can hurt fine details.

Role of densification in the sparse input case has hardly been explored.

③ Issue with Densification

- Splitting/Cloning at implausible positions → floaters
- Misplaced Gaussians can trigger further unnecessary densification
- Lack of constraints

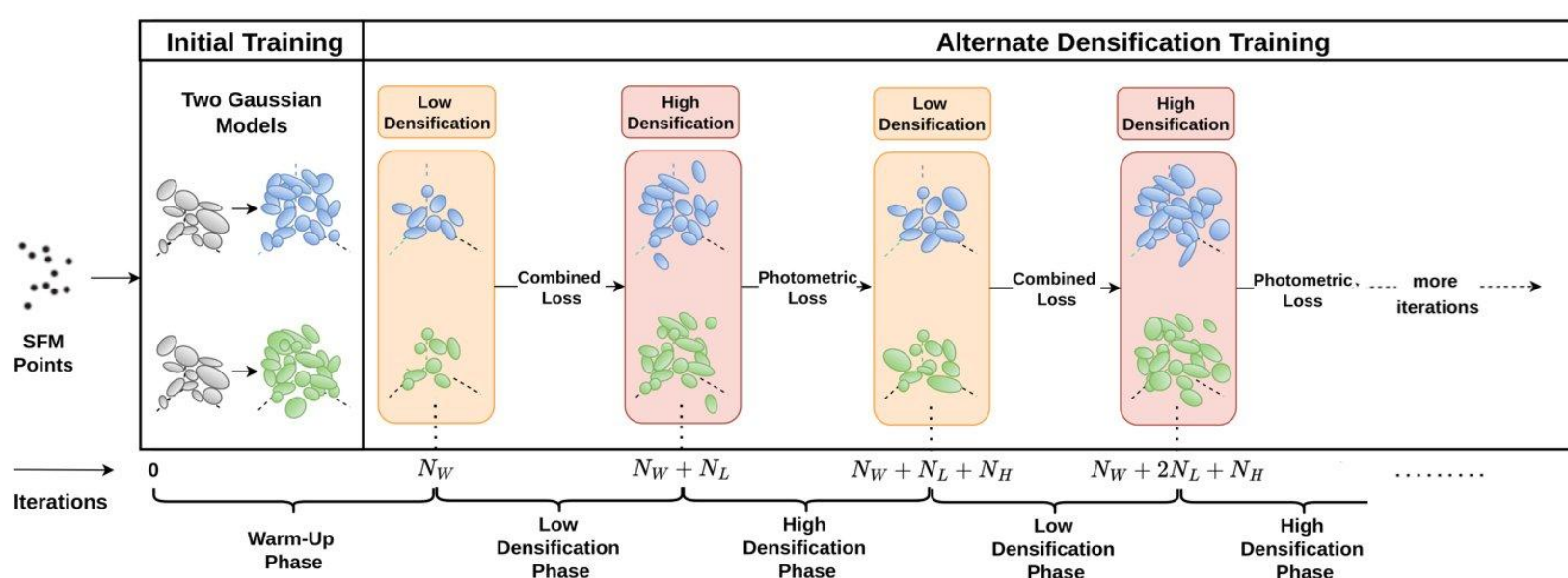
Iter 500 Iter 1000 Iter 2000 Iter 5000



- Floaters can be removed by smoothness constraints, but smoothness causes loss of details.
- Details can be recovered through aggressive densification, but end-up causing more floaters.
- Need a careful mix of smoothness and densification.

Our Idea : Alternate between Low and High Densification Phases

④ Method : AD-GS Alternating Densityfication

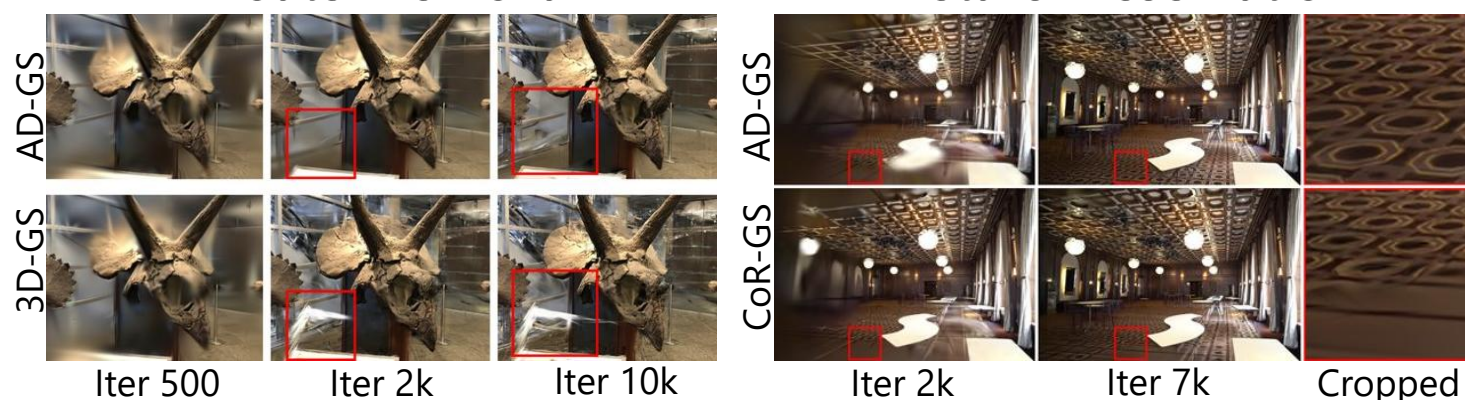


- High Densification (aggressive growth, photometric loss-only)
- Low Densification (pruning + geometry losses)
- Geometric Losses
 - Edge-Aware Depth Smoothness Loss
 - Pseudo-View Consistency Loss
- Different densification threshold and pruning threshold in each stage.

⑤ Results

Floater Removal

Details Preservation



AD-GS able to suppress floaters as well as reconstructs finer details better.

24 views (Mip-Nerf 360 Dataset)



3 views (Tanks&Temple Dataset)



Acknowledgements:

- This work was supported in part by the Kotak IISc AI-ML Centre (KIAC).

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